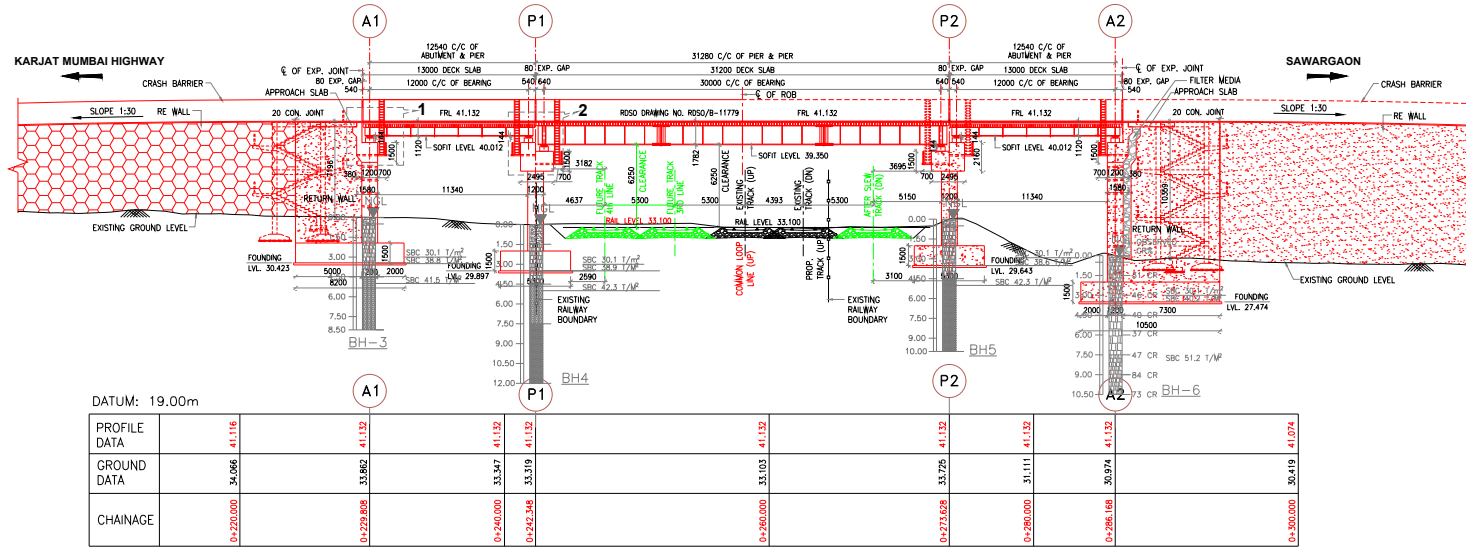
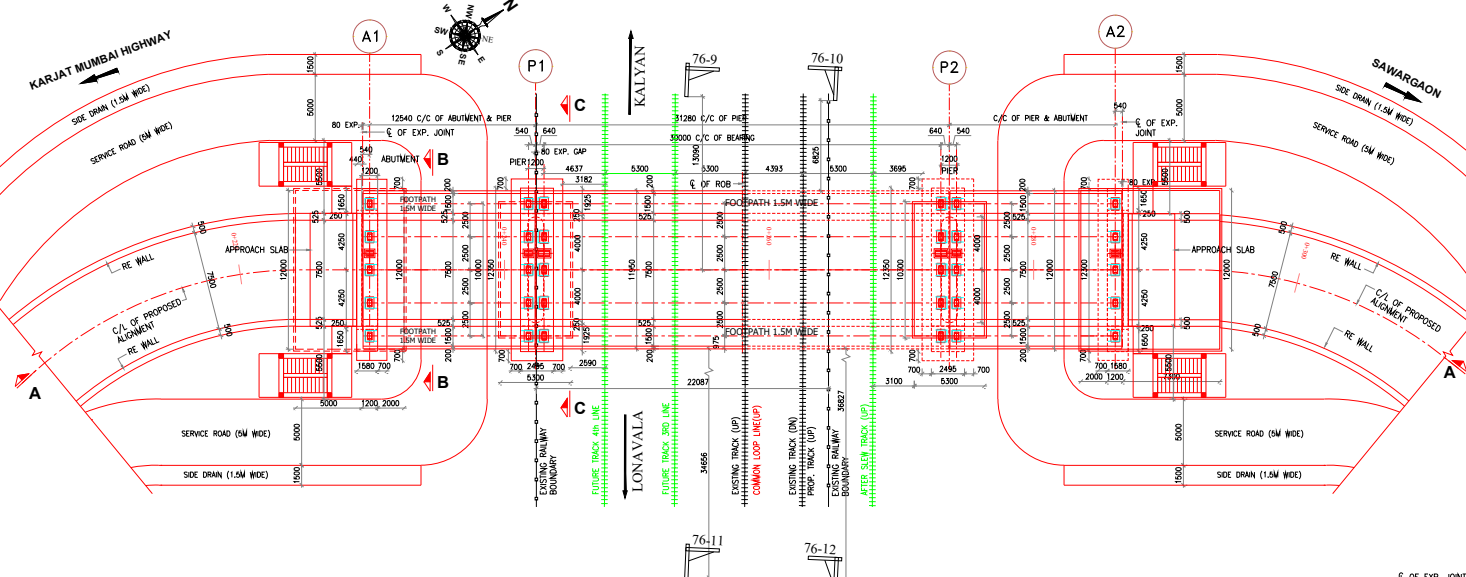
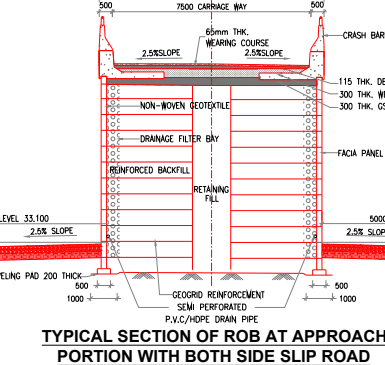




HALF ELEVATION & HALF SECTIONAL A-A
(Along C/L of Proposed Alignment)



HALF TOP & HALF BOTTOM PLAN



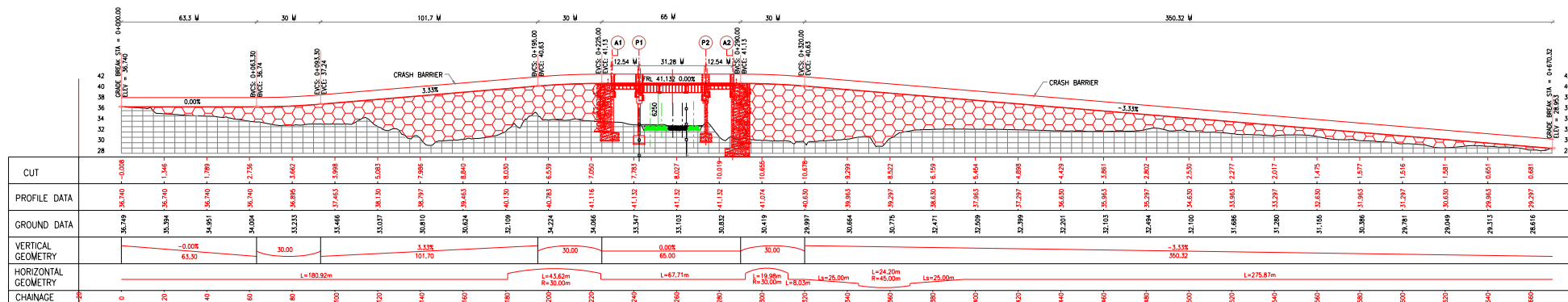
TYPICAL SECTION OF ROB AT APPROACH
PORTION WITH BOTH SIDE SLIP ROAD

DETAIL - 1

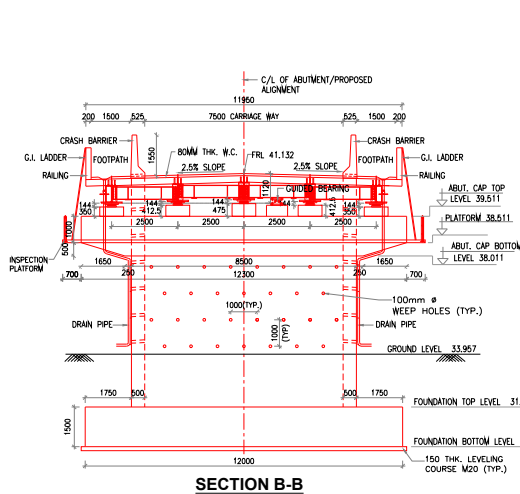
DETAIL - 2

DETAIL OF ABUTMENT CAP A1

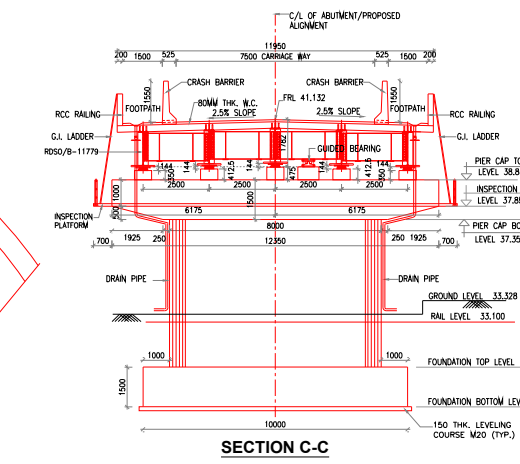
DETAIL OF PIER CAP P1



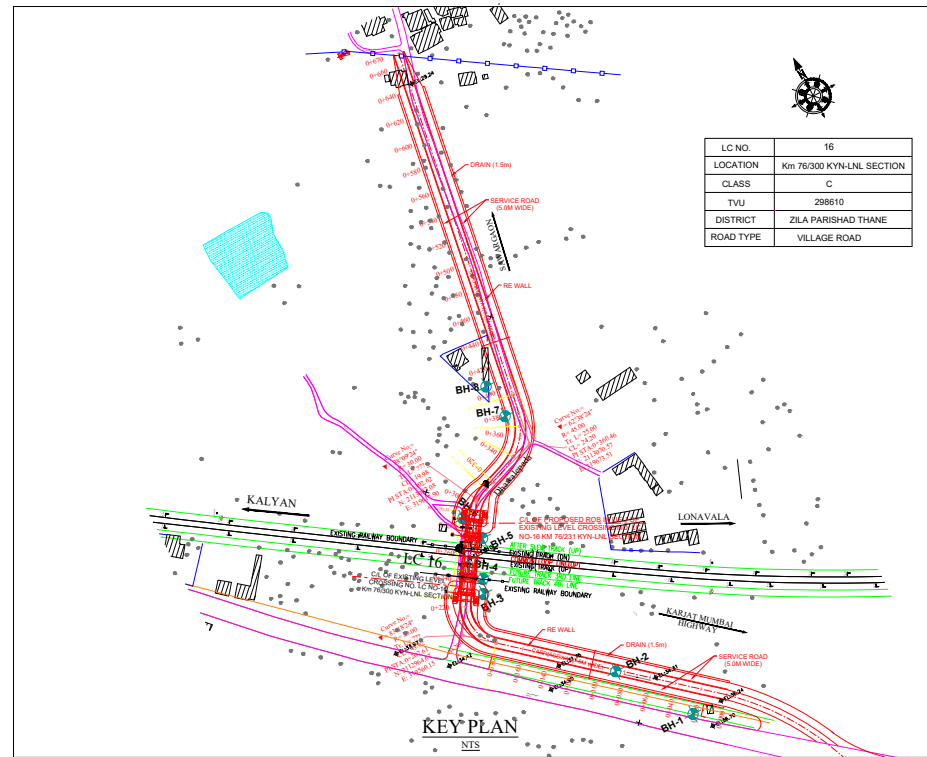
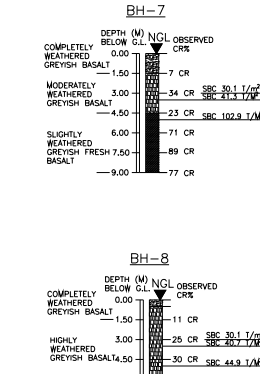
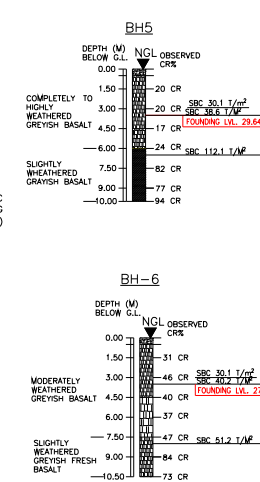
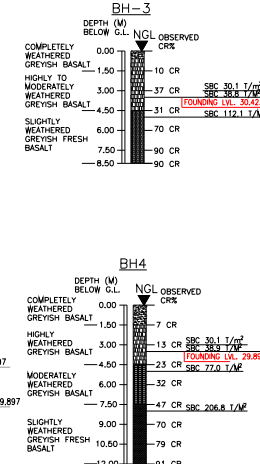
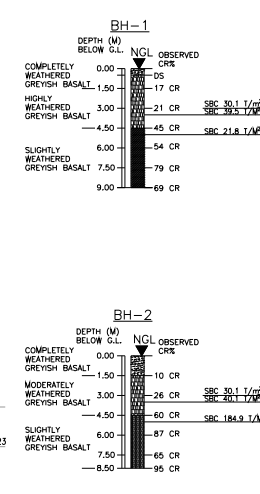
LONGITUDINAL SECTION
(SCALE: HORIZ. 1:2000 & VERT. 1:4000)



SECTION B-B



SECTION C-C



- NOTES:
1. ALL DIMENSIONS ARE SHOWN IN MM. LEVELS ARE IN METER & CHAINAGE ARE IN KILOMETERS UNLESS OTHERWISE MENTIONED.
 2. NO DIMENSIONS SHALL BE SCALED FROM DRAWING. ONLY WRITTEN DIMENSIONS TO BE FOLLOWED.
 3. ALL THE DIMENSIONS SHOWN SHOULD BE VERIFIED AT SITE BEFORE TAKING UP DESIGN WORK IN HAND. ANY DISCREPANCY NOTED SHOULD BE FORWARDED THROUGH ALTERATION DAILY SHEET.
 4. SUITABLE SPEED RESTRICTION IF REQUIRED, SHOULD BE IMPOSED BEFORE TAKING UP THE EXECUTION OF WORK WITH THE APPROVAL OF DIVISIONAL ENGINEER IN CHARGE.
 5. DURING THE CONSTRUCTION OF THE ROB, ALL EXISTING STRUCTURES AND TRACKS SHALL BE PROTECTED BY PROVIDING PROPER SHORING ARRANGEMENTS IF REQUIRED, WITH THE APPROVAL OF THE DIVISIONAL ENGINEER IN CHARGE.
 6. REPRESENTATIVE OF SIGNALING & TELECOM DEPARTMENT SHOULD BE AVAILABLE AT SITE DURING EXECUTION WITHIN VICINITY OF SIGNALING/TELECOM CABLES.
 7. ALL THE PREFABRICATED STEEL GIRDERS SHALL BE LAUNCHED USING A SUITABLE LAUNCHING SCHEME AS PER SITE REQUIREMENTS. WITH THE APPROVAL OF THE RAILWAY AUTHORITY.
 8. TEMPORARY DIVERSION LC SHALL BE PROOF CHECKED FROM REPUTED CONSULTANTS (I/T) & DESIGN INSTITUTES BEFORE EXECUTION.
 9. FOR INSPECTION OF RAILWAY SPAN, PROVISION OF INSPECTION PLATFORM FOR BOTH SIDE OF PERABUTMENT WITH RAILING AND LADDER SHOULD BE MADE.
 10. DEPTH OF FOUNDATION MAY BE ALTERED BY ENGINEER IN CHARGE TO SUIT FOUNDING STRATA MET AT SITE AS PER BEARING CAPACITY.
 11. NO PLASTERING AND CEMENT WORK PROTECTIVE COATING OVER THE CONCRETE IS PERMITTED. ANY BAD QUALITY WORK WILL BE SUMMARILY REJECTED.
 12. THE VISIBILITY OF THE SIGNAL SHALL NOT BE OBSTRUCTED.
 13. WEED HOLES AT 100MM C/C BOTH HORIZONTALLY AND VERTICALLY STAGGERED SHALL BE PROVIDED FOR APPROACH SECTION & RETAINING WALL.
 14. CRASH BARRIER SHALL BE PROVIDED AS PER HO LETTER NO. W-293 BR. CIRCULAR DATED 20.01.2023.
 15. TVU OF LC IS 288610 OF YEAR NOVEMBER 2022 (ENGG. INTERLOCKED).
 16. SAFETY, RELIABILITY AND LONG TERM DURABILITY OF THE BRIDGE SHALL BE ENSURED DURING THE DESIGN AND EXECUTION STAGES AS PER CENTRAL RAILWAY LETTER NO. W-41 BR POLICY DATED 10.10.2018.
 17. WHILE SUBMITTING THE DESIGN OF ROB FOR APPROVAL TO HEADQUARTERS, GUIDELINES ISSUED BY HQ ORS, OFFICE VIDE LETTER NO. W-2468 BR PROOF CHECKING DATED 27.08.2018 RAILWAY CROSSING STRUCTURE SHALL BE CARRIED OUT BY THE OWNER.
 18. UTILITIES IN RAILWAY PORTION AND APPROACHES ARE REQUIRED TO BE SHIFTED DURING EXECUTION.
 19. A SITE OFFICE AT THE BRIDGE SITE OR AT A NEARBY LOCATION, AS DECIDED BY THE RAILWAY, SHALL BE PROVIDED AND MAINTAINED DURING THE CONSTRUCTION PERIOD OF THE ROB.
 20. VERTICAL CLEARANCE FROM HIGHEST RAIL LEVEL TO BOTTOM LEVEL OF GROUND SHALL NOT BE LESS THAN 620MM.
 21. EXECUTION OF ANY WORK WITHIN RAILWAY BOUNDARY FOR CONSTRUCTION OF ROB SHALL BE DONE ONLY IN PRESENCE OF RAILWAYS AUTHORIZED REPRESENTATIVE TO ENSURE THE SAFETY OF RUNNING TRAINS. NO HEAVY MACHINERY WILL WORK ALONG THE TRACK WITHOUT THE PRESENCE OF RAILWAY AUTHORITY.
 22. XPM PROTECTION FRAMES SHALL BE PROVIDED FOR ELECTRIFIED SECTION AND FIXED WITH PARASUT OF ROB ON EACH LINE ON BOTH SIDES OF ROB.
 23. GALVANIZED SACRIFICIAL SHUTTERING TO BE USED FOR CASTING OF DECK SLAB, SHAPE AND SIZE OF STEEL PLATE GIRDER, CROSS GIRDERS, BEAMS, PEDESTAL, PERABUTMENT, PERABUTMENT CAP AND FOUNDATION SHOWN IN THIS DRAWING ARE TENTATIVE AND MAY VARY IN FINAL EXECUTION AFTER APPROVAL OF DESIGN BY COMPETENT AUTHORITY.
 24. DESIGN SPEED IS 250KMPH OVER ROB.
 25. A SEPARATE AGREEMENT BETWEEN RLY & ROAD AUTHORITY WILL BE EXECUTED FOR FUTURE MAINTENANCE OF ROB & SURFACE.
 26. COMPLETE ROB IS BEING CONSTRUCTED BY RAILWAY AUTHORITY AT OWN COST.
 27. AFTER COMPLETION RAILWAYS LIABILITY IS TO MAINTAIN RAILWAY PORTION SPAN 1 & FROM A-P1-P2-P3 & APPROACH PORTION WILL BE MAINTAINED BY PWD/RD AUTHORITY.
 28. APPROXIMATE AREA OF LAND ACQUISITION REQUIRED FOR THE PROPOSED SCHEME IS 1486.61 SQ.M. LAND ACQUISITION WILL BE CARRIED OUT BY RAILWAY.
 29. FABRICATION SHALL BE DONE BY ROSO APPROVED FIRM AND SHALL BE DONE IN ACCORDANCE WITH THE RAILWAY BOARD'S LETTER NO. 2017/MS-1/BRIDGE-1 DATED 20.03.2018.
 30. WORK SHALL BE CARRIED OUT IN THE FOLLOWING PHASES:
 - 31.1 TEMPORARY DIVERSION SHALL BE EXECUTED AND LIFTING BARRIERS SHALL BE SHIFTED TO CREATE TEMPORARY LC.
 - 31.2 EXISTING LC SHALL BE CLOSED.
 - 31.3 ROB SHALL BE CONSTRUCTED ALONG WITH SERVICE ROAD CONNECTIONS AND COMMISSIONED.
 - 31.4 TEMPORARY LC SHALL BE CLOSED.
 32. LC SHALL BE CLOSED AND WORK SHALL BE STARTED ONLY AFTER OBTAINING NOC FROM THE DISTRICT COLLECTOR'S OFFICE AND COMPLETION OF ROB CONSTRUCTION.
 33. STATE APPROVAL HAS BEEN OBTAINED FOR THE ROB.
 34. FOUNDATION & STRUCTURAL DETAILS ARE TENTATIVE AND SHALL BE PROVIDED AS REQUIRED FROM GEOTECHNICAL & STRUCTURAL DESIGN POINT OF VIEW. HOWEVER, CLEARANCES INDICATED SHALL NOT BE REDUCED.
 35. SUITABLE UTILITY CUT MAY BE PROVIDED WITHIN THE CRASH BARRIER TO CATER THE UTILITY SERVICE AS PER CL. 100.1 OF IRC-5:2015. HOWEVER, CABLE CROSSING OVER RAILWAY WILL REQUIRE SEPARATE APPROVAL.
 36. SAT CABLES AND GEARS TO BE PROTECTED & UTILITY SHIFTING WILL BE DONE BY SAT EXECUTING DEPARTMENT.
 37. CONSTRUCTION OF SAT WORKS WILL BE CARRIED OUT BY OPEN LINE AND NECESSARY FUND WILL BE ARRANGED BY EXECUTING AGENCY.
 38. FOR FABRICATION OF STEEL GIRDERS, FOLLOW RSO GUIDELINES BS-110, REVISED 112 AND SPECIFICATION OF BS-1200.
 39. SAND BLASTING OF STEEL GIRDERS MEMBERS METALIZING AND PROTECTIVE COATS OF APPROVED QUALITY PAINTS AS PER IS 1868 OF PARA 218.
 40. PROOF LOAD TEST AS PER IRC SP 91:2015 SHOULD BE CONDUCTED AFTER COMMISSIONING OF ROB.
 41. NEOSPHERE BEARING IS ADOPTED WITH GUARANTEE CERTIFICATE.
 42. REINFORCEMENT BARS SHALL BE H.Y.S.D. OF GRADE F460, CONFORMING TO IS 1786.
 43. STRIP SEAL TYPE EXPANSION JOINTS SHALL BE PROVIDED AS PER M.O.S.T. REQUIREMENT.
 44. EXPANSION JOINT SHALL BE PROVIDED AS PER LATEST MORTHIN O.S.T. REQUIREMENT. CONTRACTOR SHOULD FURNISH DESIGN DRAWING OF EXPANSION JOINT APPROVED BY MOST AT VARIOUS RELEVANT AMBIENT JOINT TEMPERATURE TO SUIT SITE CONDITIONS WITH THE APPROVAL OF ENGINEER IN CHARGE.
 45. DESIGN & DRAWINGS SHALL BE PROOF CHECKED FROM REPUTED CONSULTANTS (I/T) & DESIGN INSTITUTES BEFORE EXECUTION.
 46. THE DESIGNS HAVE BEEN DONE FOR TURNWHEELS AND A-CLASS LIVE LOADS WITH APPROPRIATE CONGESTION FACTOR AS PER IRC-5:2015 AND FOR SPECIAL 38ST VEHICLE GIVEN IN IRC-5:2017.
 47. CONCRETE GRADE:
 - 1. FOUNDATION, PERABUTMENTS - M35
 - 2. PER CAPABUTMENT CAP, DECK SLAB, CRASH BARRIER, PEDESTAL - M40
 - 3. PCC LEVELLING COURSE - M30
 - 4. RETAINING WALL - M35
 48. IN ORDER TO OFFER ADEQUATE RESISTANCE AGAINST CORROSION THE REINFORCEMENT BARS SHALL BE PROVIDED WITH SUITABLE PROTECTIVE COATING DEPENDING UPON ENVIRONMENTAL CONDITIONS AS PER PARA 7.1.5 OF CONCRETE BRIDGE CODE CORRECTION SLIP NO. 12 DATED 28.08.2020. AS PER RLY BOARD'S LETTER NO. 2016/MS-1/BRIDGE-1 SEMINAR 08.07.2016, HIGH STRENGTH DEFORMED STAINLESS STEEL REINFORCEMENT AS PER IS: 19651-2017 SHOULD BE USED. IF THE ROB FALLS WITHIN 300M FROM COASTAL AREA.
 49. BACK-FILLING MATERIAL BEHIND ABUTMENT FOR APPROACH SECTION SHOULD BE PROVIDED AS PER CLAUSE 7.5 OF IRS BRIDGE SUB STRUCTURE & FOUNDATION CODE. 600 MM FILTER MEDIA TO BE PROVIDED BEHIND THE RETURN WALL AND UP TO THE CUTTING LEVEL ON THE OTHER SIDE OF THE RETURN WALL.
 50. SECTION B-B IS ALSO APPLICABLE FOR ABUTMENT A3 AND SECTION C-C IS ALSO APPLICABLE FOR PIER P2 WITH LEVELS AS PER SECTIONAL ELEVATION A-A.
 51. GAD IS PREPARED AS PER APPROVED PAVE VIDE LETTER NO. W-641 BR CAOC/VINIS DATED 19.12.2024.
 52. DESIGN SHALL COMPLY WITH LATEST ISIRIS CODES: IRS: 5 (2015), 6 (2017), 21 (2020), 22 (2015), 24 (2010), 79 (2014), 112 (2011), 83 PART II (2018).
 53. APPROVAL OF GAD BY CSE OFFICE IS ONLY FOR SPAN CONFIGURATION, TYPE OF SUPERSTRUCTURE, SUBSTRUCTURE AND FOUNDATIONS. THE SIZE AND DIMENSION OF SUBSTRUCTURES AND FOUNDATIONS ARE INDICATIVE ONLY AND IT SHOULD BE DESIGNED, OPTIMIZED AND APPROVED BY COMPETENT AUTHORITY (ON CAOC) OFFICE.
 54. POE SANCTION SHALL BE OBTAINED BEFORE COMMENCEMENT OF THE WORK.
 55. IT SHOULD BE ENSURED THAT THE PROPOSED BRIDGE AND APPROACH ROADS ARE CONVENIENT AND ACCESSIBLE FOR THE PERSONS WITH DISABILITIES AS PER UNDER SECTION 40 (1) & (4) OF THE RIGHTS OF PERSONS WITH DISABILITIES ACT, 2016.
 56. APPROXIMATE AREA OF LAND ACQUISITION REQUIRED FOR THE PROPOSED SCHEME IS 1486.61 SQ.M. LAND ACQUISITION WILL BE CARRIED OUT BY RAILWAY.
 57. FABRICATION SHALL BE DONE BY ROSO APPROVED FIRM AND SHALL BE DONE IN ACCORDANCE WITH THE RAILWAY BOARD'S LETTER NO. 2017/MS-1/BRIDGE-1 DATED 20.03.2018.
 58. WORK SHALL BE CARRIED OUT IN THE FOLLOWING PHASES:
 - 31.1 TEMPORARY DIVERSION SHALL BE EXECUTED AND LIFTING BARRIERS SHALL BE SHIFTED TO CREATE TEMPORARY LC.
 - 31.2 EXISTING LC SHALL BE CLOSED.
 - 31.3 ROB SHALL BE CONSTRUCTED ALONG WITH SERVICE ROAD CONNECTIONS AND COMMISSIONED.
 - 31.4 TEMPORARY LC SHALL BE CLOSED.

LEGEND:-

LEGENDS:-

1. EXISTING WORK SHOWN IN (BLACK)
2. PROPOSED WORK SHOWN IN (RED)
3. EXISTING TRACK SHOWN IN (BLACK)
4. PROPOSED TRACK SHOWN IN (RED)
5. FUTURE 3RD & 4TH TRACK SHOWN IN (GREEN)
6. EXISTING RAIL BOUNDARY (BLACK)
7. PROPOSED RAIL BOUNDARY (BLUE)
8. BORE HOLES LOCATION SHOWN AS

APPROVAL OF GENERAL ARRANGEMENT DRAWING	
HQ APPROVAL	
DESIGNATION	SIGNATURE
CSE	DEEPAK NARAYAN KATE
DY.CE BR-I	ROHIT TENDOLKAR THANE
SSE-ROB-RUB-IV	Arjun Nigam Colaba
DIVISION APPROVAL	
DM	Digitally signed by NRESH WNA Date: 2025.06.24 20:44:53 +05'30'
ADM	Digitally signed by TRISHU KUMAR Date: 2025.06.24 19:45:00 +05'30'
SR.DM(CO)	Digitally signed by ADISH DODIA PATANWA Date: 2025.06.23 19:45:00 +05'30'
SR.DEN(CO)	Digitally signed by SHASHI LA Date: 2025.06.23 19:45:00 +05'30'
SR.DEN(CO)	Digitally signed by DEEPAK SINGH Date: 2025.06.23 19:45:00 +05'30'
SR.DEN(CO)	Digitally signed by DEEPAK SINGH Date: 2025.06.23 19:45:00 +05'30'
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SR.DEN(CO)	Digitally signed by DEEPAK SINGH Date: 2025.06.23 19:45:00 +05'30'
SR.DEN(CO)	Digitally signed by DEEPAK SINGH Date: 2025.06.23 19:45:00 +05'30'
CONSTRUCTION APPROVAL	
CE/C/RSR	Digitally signed by NRESH WNA Date: 2025.06.24 20:44:53 +05'30'
DY.CE/C/CC	Digitally signed by TRISHU KUMAR Date: 2025.06.24 19:45:00 +05'30'
DY.CE/C/RSR	Digitally signed by SHASHI LA Date: 2025.06.23 19:45:00 +05'30'
AE/C/RSR	Digitally signed by DEEPAK SINGH Date: 2025.06.23 19:45:00 +05'30'
SSE/DMD/C/RSR	Digitally signed by DEEPAK SINGH Date: 2025.06.23 19:45:00 +05'30'
CONSULTANT - RITES Ltd.	
DESIGNATION	SIGNATURE
APPROVED BY	JAM/CEDS
CHECKED BY	JAM/CEDS
DESIGN BY	JAM/CEDS
PREPARED BY	TECH. GRADE II
LOADING STANDARD	
IRC-8 2017	
SEISMIC ZONE - III	
GENERAL ARRANGEMENT DRAWING	
DIVISION	MUMBAI
SECTION	KALYAN-LONAVALA (KYN-LNL)
EXECUTIVE AGENCY	RSP UNIT, CENTRAL RAILWAY
CONSULTANT	RITES Limited (A Government of India Enterprise) RITES Bhamburda, Plot No. 1, Sector-20 Gurgaon-122001, Haryana (India)
PROJECT	DETAILED DESIGN CONSULTANCY SERVICES FOR CONSTRUCTION OF 16 ROAD OVER BRIDGES, GATI SHAKTI UNIT OF MUMBAI DIVISION OF CENTRAL RAILWAY. PB ITEM NO. 216 OF 2020-21
DRAWING TITLE	CONSTRUCTION OF ROB AT KM: 76/231.00 IN LIEU OF LC-16 AT KM: 76/300 BETWEEN BADAPUR-VANGANI ON KALYAN-LONAVALA SECTION (ON KARJAT MUMBAI HIGHWAY-SAWARGAON ROAD) SPAN: 1x12+1x30m+1x12 C.G.
DRAWING NO.	CONSULTANT: RITES/JAM/CEDS/RSP-MUMBAI/ROB-16/GAD. CONSTRUCTION: GME/RSR-1001-ROB HEAD QUARTER:
SCALE:	1:200 & 1:100